PURGE: Projected Unlearning via Retain-Guided Erasure Supplementary Material

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This document provides supplementary material for the main paper: full theorem proofs, background and preliminaries, extended experimental results, and additional analyses.

1 Background and Preliminaries

Definition 1 (Retain-Safe Update)**.** *A parameter update $\theta \to \theta - \eta g$ is* retain-safe *if it does not increase the retain loss in the first-order approximation:*

$$\langle g, \nabla_\theta \mathcal{L}_{\mathrm{r}}(\theta) \rangle \geq 0 \tag{1}$$

where $\mathcal{L}_{\mathrm{r}}(\theta) = \frac{1}{|\mathcal{D}_r|} \sum_{(x,y)\in\mathcal{D}_r} \mathrm{CE}(f_\theta(x), y)$.

1.1 A-GEM for Continual Learning

Averaged Gradient Episodic Memory (A-GEM) (Chaudhry et al. 2019) solves:

$$\min_g \|g - g_{\text{new}}\|^2 \quad \text{s.t.} \quad \langle g, g_{\text{old}} \rangle \geq 0 \tag{2}$$

where g_{new} is the new-task gradient and g_{old} is the old-task reference gradient. When $\langle g_{\text{new}}, g_{\text{old}} \rangle < 0$ (conflict), the closed-form solution is:

$$\tilde{g} = g_{\text{new}} - \frac{\langle g_{\text{new}}, g_{\text{old}} \rangle}{\|g_{\text{old}}\|^2} g_{\text{old}} \tag{3}$$

which projects g_{new} onto the boundary of the half-space $\{g : \langle g, g_{\text{old}} \rangle \geq 0\}$.

1.2 CL–MU Duality

In continual learning, A-GEM ensures that learning new tasks does not increase the loss on old tasks. In machine unlearning, we repurpose this: we ensure that *forgetting* specific data does not increase the loss on *retained* data. The mapping is:

Component	CL (A-GEM)	MU (PURGE)
Primary gradient	New-task gradient	Forget gradient g_{f}
Reference gradient	Old-task gradient	Retain gradient g_{r}
Constraint	Don't increase old loss	Don't increase retain loss
Projection trigger	$\langle g_{\text{new}}, g_{\text{old}} \rangle < 0$	$\langle g_{\mathrm{f}}, g_{\mathrm{r}} \rangle < 0$

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2 Full Proofs

2.1 Proof of Theorem 1 (Retain-Safety)

Theorem 1 (Retain-Safety with KD, Clipping, and Momentum)**.** *Let the composite retain objective be $\widetilde{\mathcal{L}_{\mathrm{r}}}(\theta) = \mathcal{L}_{\mathrm{r}}(\theta) + \beta\, \mathcal{L}_{\mathrm{KD}}(\theta, \theta_0; \mathcal{D}_r)$, and define the reference retain gradient $g_{\mathrm{r}} := \nabla_\theta \widetilde{\mathcal{L}_{\mathrm{r}}}(\theta)$. Let $\tilde{g}_{\mathrm{f}}$ be the projected forget gradient. Define the clipped, momentum-augmented update as*

$$v_{t+1} = \mu\, v_t + \tilde{g}_{\mathrm{f}}^{(t)}, \tag{4}$$

$$u_{t+1} = \mathrm{clip}_c(v_{t+1}) := v_{t+1} \cdot \min\Big(1,\, \tfrac{c}{\|v_{t+1}\|}\Big), \tag{5}$$

$$\theta_{t+1} = \theta_t - \eta\, u_{t+1}, \tag{6}$$

where $\eta > 0$, $\mu \in [0, 1)$, $c > 0$. Assume $\widetilde{\mathcal{L}_{\mathrm{r}}}$ is L-smooth. Then:

$$\widetilde{\mathcal{L}_{\mathrm{r}}}(\theta_{t+1}) \leq \widetilde{\mathcal{L}_{\mathrm{r}}}(\theta_t) + \eta\, \mu\, \|g_{\mathrm{r}}^{(t)}\|\, \|v_t\| + \tfrac{1}{2} L\, \eta^2 c^2. \tag{7}$$

Proof. By L-smoothness:

$$\widetilde{\mathcal{L}_{\mathrm{r}}}(\theta_{t+1}) \leq \widetilde{\mathcal{L}_{\mathrm{r}}}(\theta_t) + \langle g_{\mathrm{r}}^{(t)}, \theta_{t+1} - \theta_t \rangle + \tfrac{L}{2} \|\theta_{t+1} - \theta_t\|^2.$$

Second-order term. $\|\theta_{t+1} - \theta_t\|^2 = \eta^2 \|u_{t+1}\|^2 \leq \eta^2 c^2$ by the clipping definition, giving the second-order term $\frac{1}{2} L \eta^2 c^2$.

First-order term.

$$\begin{aligned} \langle g_{\mathrm{r}}^{(t)}, \theta_{t+1} - \theta_t \rangle &= -\eta\, \langle g_{\mathrm{r}}^{(t)}, u_{t+1} \rangle \\ &= -\eta\, \alpha\, \langle g_{\mathrm{r}}^{(t)}, v_{t+1} \rangle, \end{aligned}$$

where $\alpha := \min(1, c/\|v_{t+1}\|) \in (0, 1]$ is the clipping factor.

Expanding $v_{t+1} = \mu v_t + \tilde{g}_{\mathrm{f}}^{(t)}$:

$$\langle g_{\mathrm{r}}^{(t)}, v_{t+1} \rangle = \mu\, \langle g_{\mathrm{r}}^{(t)}, v_t \rangle + \underbrace{\langle g_{\mathrm{r}}^{(t)}, \tilde{g}_{\mathrm{f}}^{(t)} \rangle}_{\geq 0 \text{ by projection}}.$$

The projected term is non-negative by A-GEM: if $\langle g_{\mathrm{f}}, g_{\mathrm{r}} \rangle \geq 0$ then $\tilde{g}_{\mathrm{f}} = g_{\mathrm{f}}$ and $\langle g_{\mathrm{f}}, g_{\mathrm{r}} \rangle \geq 0$; if $\langle g_{\mathrm{f}}, g_{\mathrm{r}} \rangle < 0$ the projection gives $\langle \tilde{g}_{\mathrm{f}}, g_{\mathrm{r}} \rangle = 0$ (a direct computation). Hence:

$$\langle g_{\mathrm{r}}^{(t)}, \theta_{t+1} - \theta_t \rangle \leq -\eta\, \alpha\, \mu\, \langle g_{\mathrm{r}}^{(t)}, v_t \rangle \leq \eta\, \mu\, \|g_{\mathrm{r}}^{(t)}\|\, \|v_t\|$$

by Cauchy–Schwarz and $\alpha \leq 1$. Combining with the second-order term yields the bound. □

Remark 1 (Role of each hyperparameter)**.** • η *(learning rate): both terms shrink as* $\eta \to 0$*; the second-order term is* $O(\eta^2)$*.*

- μ *(momentum): if* $\mu = 0$ *the leakage term vanishes entirely; with* $\mu = 0.9$ *it is the main source of deviation from the pure-projection guarantee.*
- c *(clip norm): bounds the second-order term and bounds* $\|v_t\|$ *recursively. Small* c *tightens the bound but slows convergence.*
- β *(KD weight): does* not *appear in the bound because the projection is computed against the composite retain gradient* $g_{\mathrm{r}} := \nabla\widetilde{\mathcal{L}_{\mathrm{r}}}$*, which already contains* $\beta\nabla\mathcal{L}_{\mathrm{KD}}$*.*

Corollary 2 (A-GEM Reduction)**.** *Setting* $\mu = 0$*,* $\beta = 0$*,* $c = \infty$ *recovers the original A-GEM guarantee:* $\mathcal{L}_{\mathrm{r}}(\theta_{t+1}) \le \mathcal{L}_{\mathrm{r}}(\theta_t) + O(\eta^2)$*.*

2.2 Proof of Theorem 2 (Per-Step Separation)

Theorem 3 (Per-Step Retain-Safety Separation)**.** *Fix a model* θ_0*, forget set* $\mathcal{D}_f$*, retain set* $\mathcal{D}_r$ *with retain gradient* $g_{\mathrm{r}} := \nabla\mathcal{L}_{\mathrm{r}}(\theta_0)$*. Consider one-step update operators for GA, SCRUB, SalUn, and* PURGE*. Then:*

1. ***GA.*** *The update* $\theta_0 + \eta g_{\mathrm{f}}$ *strictly increases retain loss when* $\langle g_{\mathrm{f}}, g_{\mathrm{r}}\rangle > 0$*.*
2. ***SCRUB.*** *With weighting* $\lambda > 0$*, the combined direction* $-g_{\mathrm{f}} + \lambda g_{\mathrm{r}}$ *can have* $\langle \cdot, g_{\mathrm{r}}\rangle < 0$ *for any fixed* λ*.*
3. ***SalUn.*** *The masked random-label gradient* $-M \odot \nabla\mathrm{CE}_{\mathrm{RL}}$ *has unconstrained sign w.r.t.* g_{r}*.*
4. **PURGE.** $\langle \tilde{g}_{\mathrm{f}}, g_{\mathrm{r}}\rangle \ge 0$ *unconditionally.*

Proof. **(1) GA.** $\mathcal{L}_{\mathrm{r}}(\theta_0 + \eta g_{\mathrm{f}}) = \mathcal{L}_{\mathrm{r}}(\theta_0) + \eta\langle g_{\mathrm{f}}, g_{\mathrm{r}}\rangle + O(\eta^2)$, strictly larger than $\mathcal{L}_{\mathrm{r}}(\theta_0)$ when $\langle g_{\mathrm{f}}, g_{\mathrm{r}}\rangle > 0$.

(2) SCRUB. $\langle -g_{\mathrm{f}} + \lambda g_{\mathrm{r}}, g_{\mathrm{r}}\rangle = -\langle g_{\mathrm{f}}, g_{\mathrm{r}}\rangle + \lambda\|g_{\mathrm{r}}\|^2$. Choose any $g_{\mathrm{f}}, g_{\mathrm{r}}$ with $\langle g_{\mathrm{f}}, g_{\mathrm{r}}\rangle > \lambda\|g_{\mathrm{r}}\|^2$ (satisfiable whenever $\langle g_{\mathrm{f}}, g_{\mathrm{r}}\rangle/\|g_{\mathrm{r}}\|^2 > \lambda$, which holds in any neighbourhood where forget and retain gradients are positively correlated). Then the inner product is negative.

(3) SalUn. With random labels, the masked gradient direction is uncorrelated with g_{r} in expectation; specific draws produce arbitrary sign for $\langle g_{\mathrm{SalUn}}, g_{\mathrm{r}}\rangle$.

(4) PURGE. Follows from Theorem 1 and the A-GEM projection definition. □

Remark 2. *Theorem 2 states a* worst-case *separation: there exist configurations where GA, SCRUB, or SalUn strictly degrade retain loss in a single step, whereas* PURGE *provably does not. In practice, all methods often produce small retain-loss changes; the point is that only* PURGE guarantees *this by construction.*

3 Empirical Findings (Extended)

3.1 Finding 1: MIA AUROC is Insufficient

Finding 1. *For class-level unlearning on well-generalising classifiers, AUROC-based MIA is not a sufficient privacy metric and must be paired with forget accuracy (FA).*

Evidence.

1. *Base-model control*: the pre-unlearning ResNet-18 on CIFAR-10 has MIA AUROC $= 0.533$ at FA $= 99.7\%$. The model fully recognises the forget class, yet MIA is near random.
2. *Methods that barely forget also reach MIA* ≈ 0.5. SCRUB (FA $= 88.8\%$, MIA $= 0.515$) and NegGrad+ (FA $= 84.8\%$, MIA $= 0.525$).
3. *Methods that genuinely forget*: SalUn (FA $= 0.0$, MIA $= 0.497$), PURGE (FA $= 2.1$, MIA $= 0.530$), Bad Teacher (FA $= 29.7$, MIA $= 0.499$).

Conditional bound. If, after unlearning, $\mathrm{KL}(p^*_\theta(x_f)\|p_r) \le \epsilon$ for all $x_f \in \mathcal{D}_f$, then by Pinsker's inequality $\mathrm{TV}(p^*_\theta(x_f), p_r) \le \sqrt{\epsilon/2}$, and the AUROC of any output-observing MIA attacker is bounded by $|\mathrm{AUROC}(A) - 0.5| \le \frac{1}{2}\sqrt{\epsilon/2}$.

3.2 Finding 2: FA Non-Monotonicity

Finding 2. *The projected* `kl_retain` *update decreases forget loss in expectation, but forget* accuracy *is not monotonically non-increasing.*

FA is piecewise-constant in θ: it changes only when $\arg\max$ flips on at least one forget sample. Between flips, FA stays exactly constant even as KL loss decreases. Stochastic mini-batch projection occasionally pushes the argmax back to the original label, transiently increasing FA.

Correct statement. Under the assumptions of Theorem 1, the *forget loss* satisfies:

$$\mathbb{E}\big[\mathrm{KL}(p_{\theta_{t+1}}(x_f)\|p_r)\big] \le \mathrm{KL}(p_{\theta_t}(x_f)\|p_r) - \eta\,\xi_t + O(\eta^2),$$

where $\xi_t \ge 0$ is the projection-preserved descent magnitude.

3.3 Finding 3: BatchNorm Vulnerability

Finding 3. *Class-homogeneous forget batches corrupt BN running statistics, manufacturing apparent forgetting that one retain forward pass reverses. Replacing BN with GroupNorm eliminates this entirely.*

BN recalibration results on CIFAR-10 class 0:

Method	Pre-F	Post-F	ΔF	Verdict
Base (no unlearn)	98.2	99.1	+0.9	weights (n/a)
SalUn	0.0	0.0	+0.0	weights
Bad Teacher	28.7	27.1	−1.6	weights
SCRUB	86.5	81.2	−5.3	weights
NegGrad+	79.7	76.6	−3.1	weights
PURGE (BN)	1.3	93.5	+92.2	BN-mediated
PURGE (GN)	3.1	3.1	+0.0	weights

Multi-dataset GroupNorm evaluation:

Dataset	TA (BN→GN)	FA (BN→GN)	RA (BN→GN)
CIFAR-10	86.0→84.1	2.1→3.1	100.0→99.1
MNIST	90.3→90.1	8.4→5.3	99.4→99.3
SVHN	89.6→87.9	5.1→2.3	97.2→95.4
STL-10	92.6→83.8	72.8→96.6	99.4→93.0
PathMNIST	73.0→71.9	0.02→2.0	98.9→95.9

GroupNorm improves CIFAR-10 and MNIST but degrades STL-10 (FA 96.6%) and PathMNIST (RA drop). We recommend GroupNorm for CIFAR-10/MNIST deployments but do not claim it as a universal fix.

3.4 Finding 4: Representation-Level Non-Erasure

Finding 4. *A linear probe on penultimate-layer features recovers the forget class at 95–98% from all approximate unlearning methods, including* PURGE*. Output-level forgetting does not imply representation-level erasure.*

Linear probe accuracy (5-fold CV logistic regression on class-0 features, CIFAR-10):

Method	FA (%)	Probe Acc. (%)
Base model	98.2	98.2 ± 0.5
SalUn	0.0	97.0 ± 0.9
Bad Teacher	29.7	94.9 ± 1.1
SCRUB	88.8	97.5 ± 0.9
NegGrad+	84.8	97.8 ± 0.3
PURGE	2.1	95.0 ± 1.5

4 Component-Isolation Ablations (Full Table)

Configuration	TA	FA↓	RA	MIA	What it tests
Full PURGE	85.2	8.9	97.2	0.507	baseline (seed 42)
−KD ($\beta = 0$)	81.4	6.5	92.7	0.510	KD retain stability
−Rep ($\lambda = 0$)	85.4	9.5	97.3	0.506	rep-erasure at output
−Stopping	81.0	0.0	92.3	0.532	retain-damage prevention
−Gate/cap	85.4	9.5	97.3	0.506	stability contribution
−BN-freeze	82.2	8.2	94.1	0.505	BN corruption
−Projection	79.7	6.0	90.8	0.511	retain safety source
Selective-layer	85.4	8.2	97.4	0.499	early-layer freezing

5 Sweep A: Retain-Loss Budget

Sweep A fixes FA_target $= 10\%$ and varies $\alpha \in \{1.5, 2.0, 3.0, 5.0, 10.0, 9999\}$ on CIFAR-10 class 0 (seed 42).

α	TA	FA↓	RA	MIA	Stop reason
1.5	85.4	9.5	97.3	0.506	FA target
2.0	85.4	9.5	97.3	0.506	FA target
3.0	85.4	9.5	97.3	0.506	FA target
5.0	85.4	9.5	97.3	0.506	FA target
10.0	85.4	9.5	97.3	0.506	FA target
9999	85.4	9.5	97.3	0.506	FA target

ϵ_R is non-binding: all runs stop at the FA target at epoch 2. The retain-loss budget acts as a safety net, not the active constraint.

6 Sequential Unlearning (Extended)

Step	Forget Cls	Forgotten	TA	FA↓	RA
1	0	{0}	86.1	2.3	100.0
2	1	{0,1}	76.2	0.3	89.1
3	2	{0,1,2}	66.9	1.2	77.8

FA remains low at each step ($\leq 2.3\%$), confirming the forget objective works. However, TA drops $\sim$10pp per step and RA degrades from 100% to 77.8% after three steps. The cumulative damage arises because each step's retain set includes previously forgotten classes.

7 Fine-Tuning Recovery Attack (Extended)

Epoch	TA	FA	RA	MIA
0 (post-PURGE)	86.1	2.2	100.0	0.531
5	79.2	11.9	97.8	0.501
10	34.2	67.3	45.0	0.442

Fine-tuning on 50 forget-class samples with $\mathrm{lr} = 10^{-3}$. After 5 epochs, FA rises to 11.9%; by epoch 10, the model fully recovers the forget class. This vulnerability is common to all approximate methods.

8 CIFAR-100 Per-Class Results

Forget Cls	TA (%)	FA (%)↓	RA (%)	MIA
10	77.0	5.0	92.6	0.487
20	76.2	1.6	92.2	0.478
30	74.0	1.0	89.8	0.492
40	76.7	9.0	92.8	0.503
50	75.2	1.2	91.6	0.555
60	76.3	17.6*	92.1	0.437
70	73.8	1.6	88.9	0.480
80	74.0	1.0	90.2	0.556
90	75.1	2.0	91.1	0.503
Mean	**75.4**	**4.4**	**91.3**	**0.499**

*Class 60 did not converge within 20 epochs.

9 Per-Class Results: MNIST (10 Classes)

Forget Cls	TA (%)	FA (%)	RA (%)	MIA
0	90.3	8.4	99.4	0.495
1	88.3	1.9	99.5	0.513
2	89.6	5.3	99.5	0.491
3	90.3	8.2	99.4	0.484
4	90.3	8.6	99.4	0.510
5	90.8	4.0	99.5	0.499
6	90.1	2.2	99.5	0.502
7	90.0	9.3	99.5	0.491
8	90.4	9.7	99.5	0.486
9	90.0	5.1	99.5	0.504
Avg	**90.0**	**6.3**	**99.5**	**0.497**

10 Per-Class Results: PathMNIST (9 Classes)

Cls	Tissue Type	TA	FA	RA	MIA
0	Adipose	73.0	0.0	98.9	0.515
1	Background	78.3	0.6	98.3	0.456
2	Debris	89.2	6.9	98.9	0.298
3	Lymphocytes	82.2	5.0	98.8	0.386
4	Mucus	78.0	8.4	98.9	0.501
5	Smooth Musc.	84.5	5.1	98.2	0.548
6	Normal Muc.	78.4	9.5	98.7	0.541
7	Stroma	88.5	6.2	99.1	0.562
8	Epithelium	75.2	9.8	99.2	0.530
	Average	**80.8**	**5.7**	**98.8**	**0.482**

11 Feature-Space Fréchet Distance

We compute the Fréchet distance between penultimate-layer features of the unlearned and retrained models on the forget

set:

$$\mathrm{FID}_{\text{feat}} = \|\mu_u - \mu_r\|^2 + \mathrm{Tr}\Big(\Sigma_u + \Sigma_r - 2(\Sigma_u\Sigma_r)^{1/2}\Big)$$

Method	$\mathrm{FID}_{\text{feat}} \downarrow$	Cos ↓	L2 ↑
PURGE (`kl_retain`)	195.7	0.742	16.37
PURGE (GA)	102.3	0.841	12.17
PURGE (KL-uni)	412.8	0.430	21.39

GA produces the lowest $\mathrm{FID}_{\text{feat}}$ because it barely modifies representations (consistent with under-forgetting). KL-uniform produces the highest, pushing toward an artificial target. `kl_retain` sits in between with the best cosine drop, modifying class-specific activations while converging toward the retain distribution.

12 Hyperparameter Configuration

Hyperparameter	CIFAR-10	Others
Learning rate (η)	5×10^{-4}	2×10^{-4}
Max epochs (T)	15	12–15
Rep. erasure wt. (λ)	0.05	0.03–0.05
KD weight (β)	2.0	3.0–5.0
Grad. clip norm (c)	1.0	1.0
Budget factor (α)	5.0	4.0–5.0
Forget gate (γ)	0 (off)	0
Distill. temp. (T_{KD})	4.0	4.0
FA check freq.	50 batches	50 batches

13 Per-Step Retain-Loss Trajectory

To empirically validate the bound in Theorem 1, we log the retain CE loss before and after every gradient step during PURGE unlearning on CIFAR-10 class 0 (seed 42, 5 epochs, 200 steps).

Key observations.

1. The per-step retain-loss delta is small in early steps ($\sim 10^{-5}$ to 10^{-4}) and grows in later epochs ($\sim 10^{-3}$), consistent with the momentum leakage term $O(\eta\mu\|v_t\|)$ in Theorem 1. As the momentum buffer v_t accumulates, the leakage increases—exactly as predicted.
2. Forget CE rises monotonically from ∼0.002 to >7.0 across 200 steps, confirming progressive erasure.
3. The cumulative retain-loss increase across 200 steps is ∼0.03, well within the retain-loss budget ($\alpha L_0 \approx 0.45$ at $\alpha = 5$). The dual stopping criterion never triggers on the retain-loss side.

These observations confirm that the theoretical bound is not vacuous in practice: the per-step retain-loss change is small and well-controlled, and the residual momentum leakage is caught by the retain-loss budget before it can accumulate to a meaningful level.

14 Algorithm-to-Theory Mapping

Algorithm 1 in the main paper can be read as a sequence of steps, each realising a specific term of the constrained optimisation problem (Eq. 1):

Table 1: Forget-objective ablation on CIFAR-10 (class 0).

Configuration	TA	FA↓	RA	MIA
Full PURGE (`kl_retain`)	84.5	8.4	96.3	0.496
GA objective	89.3	52.6	97.0	0.250
GA (more epochs)	80.8	0.0	93.7	0.243
KL-uniform	83.8	12.0	93.9	0.469
Ent. gating (γ=0.9)	89.1	58.6	96.2	0.262
BN in train mode	82.2	8.2	94.1	0.505

Steps 1–2: Teacher snapshot and BN freeze. A frozen copy of θ_0 provides the stable teacher distribution for $\mathcal{L}_{\mathrm{KD}}$. The BN freeze ensures apparent forgetting is realised in weights rather than BN running statistics (Finding 3).

Step 3: Retain-confusion estimation. Computes $p_r = \frac{1}{|\mathcal{D}_r|}\sum_{x\in\mathcal{D}_r}\sigma(\theta_0(x))$, realising the target distribution for the forget-distribution constraint $\mathrm{KL}(p_\theta(x_f)\|p_r) \le \epsilon_F$.

Step 4: Initial retain-loss budget. Computes $L_0 = \mathrm{CE}(\theta, \mathcal{D}_r)$ and the stopping threshold αL_0, realising the retain-loss constraint $\mathcal{L}_{\mathrm{r}}(\theta) \le \mathcal{L}_{\mathrm{r}}(\theta_0) + \epsilon_R$ with $\epsilon_R = (\alpha - 1)L_0$.

Steps 5–6: Retain gradient with KD. The composite retain gradient $g_{\mathrm{r}} = \nabla[\mathrm{CE} + \beta\mathcal{L}_{\mathrm{KD}}]$ provides the reference direction for the projection. Including the KD term means the per-step bound holds w.r.t. the composite retain objective.

Step 7: Entropy cap. If $\mathrm{CE}(\theta(x_f), y_f) \ge \log K$, the model is already maximally uncertain on the forget batch; the update is restricted to the retain objective, preventing gradient explosion. On gated steps the update is a pure retain step, which still satisfies Theorem 1.

Steps 8–10: Forget gradient, projection, and clipping. The central step: the projection enforces the retain-loss constraint per step (Theorem 1); clipping bounds the $O(\eta^2 c^2)$ term; momentum introduces the $O(\eta\mu\|v\|)$ leakage.

Steps 11–14: Dual stopping. Two early-exit checks: FA ≤ target (realising ϵ_F) and retain loss > budget (realising ϵ_R). The FA check is performed intra-epoch every 50 batches due to FA's piecewise-constant dynamics (Finding 2).

15 GA vs. KL-Retain vs. KL-Uniform Ablation

Table 1 compares the three forget objectives along with entropy gating and BN mode ablations on CIFAR-10 class 0.

GA vs. `kl_retain`. GA either under-forgets (FA = 52.6% after a few epochs) or, pushed further, drives FA to 0.0% at a retain cost (RA = 93.7%). In both cases MIA AUROC (∼0.25) is far from ideal—the model's outputs are clearly distinguishable from a retrained model. `kl_retain` achieves FA = 8.4% with RA = 96.3% and MIA = 0.496, near random chance on all three axes.

KL-uniform. Targeting the uniform distribution achieves FA $= 12.0\%$ and RA $= 93.9\%$, but MIA AUROC $= 0.469$—detectably below the ideal 0.5. The artificial uniform target is detectable because no retrained model produces uniform outputs; `kl_retain`'s advantage comes from mimicking the natural confusion pattern.

Entropy gating. Setting $\gamma = 0.9$ causes the gate to activate for 98% of batches, leading to severe under-forgetting (FA $= 58.6\%$). The default $\gamma = 0$ (gate off) yields a better balance.

16 Design Rationale: Why We Abandoned IEWPv2

The current PURGE design followed an earlier method, IEWPv2 (Influence-Weighted Entropy Unlearning Protocol v2), which combined influence-weighted entropy maximisation with interleaved retain training. Two fundamental limitations motivated the switch:

1. IEWPv2 relied on gradient norms as a proxy for influence scores rather than computing the true inverse-Hessian-vector product, weakening its theoretical grounding.
2. Its alternating forget–retain optimisation steps led to partial cancellation: the retain step counteracted forgetting, and vice versa, resulting in slow and unstable convergence that worsened with additional classes.

PURGE addresses both issues by jointly handling forget and retain objectives through a single projected update: the projection ensures the forget direction does not conflict with the retain direction, eliminating the cancellation problem entirely.

17 Ethical and Societal Considerations

Unlearning as evidence removal. An unlearning method that makes a model indistinguishable from one retrained without specific data could be misused to destroy evidence that problematic data was used for training. PURGE's audit trail (projection rate logs, FA trajectory, retain-loss budget history) provides partial mitigation.

Selective fairness degradation. Class-level unlearning removes all samples from a class, which may correspond to a demographic group. The TA degradation we observe is not confined to the forgotten class. Deployers should monitor per-class accuracy before and after unlearning.

Regulatory gap. Approximate unlearning methods, including PURGE, do not satisfy formal (ε, δ)-DP certification. Whether MIA AUROC ≈ 0.5 constitutes legally sufficient "erasure" remains an open legal question.

18 Reproducibility

Code, configs, and logs are publicly available. Key implementation details:

- All experiments use PyTorch ≥ 2.0 with CUDA.
- YAML config files for each dataset are provided.
- The retain confusion distribution p_r is precomputed once.
- Forward hooks are registered on `layer3`, `layer4`, and `avgpool` for ResNet-18.
- BN layers are set to `eval()` mode immediately after setting the model to `train()` mode.
- All baselines are re-run with unified code and MIA protocol.

References

Chaudhry, A.; Ranzato, M.; Rohrbach, M.; and Elhoseiny, M. 2019. Efficient Lifelong Learning with A-GEM. In *International Conference on Learning Representations (ICLR)*.